

Physical Chemistry
Topic- Atomic Structure

Work Sheet - 10

1. Complete the following table :

Q.No.	Ionization energy	2 nd excitation energy	Binding energy of 2 nd state
1.	Li^{2+}		
2.	He^+		
3.	H		

2. Complete the following table :

Q.No.	Species	Transition	Energy of emitted photons	Wavelength of emitted photons (in terms of R)
1.	H-atom	$2 \rightarrow 1$		
2.	He^+ ion	$3 \rightarrow 2$		

3. Complete the following table :

Q.No.	Initial level	Final level	Maximum Number of spectral lines
1.	7th excited state	2nd state	
2.	7th state	1st excited state	

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Work Sheet - 11

1. A golf ball has a mass of 40 g, and a speed of 45 m/s. If the speed can be measured within accuracy of 2%, calculate the uncertainty in the position. [NCERT]

2. Uncertainty in position is twice the Uncertainty in momentum Uncertainty in velocity is -

(A) $\frac{1}{m} \sqrt{\frac{h}{\pi}}$

(B) $\frac{1}{2m} \sqrt{\frac{h}{\pi}}$

(C) $\frac{1}{2m} \sqrt{\frac{h}{2\pi}}$

(D) $\frac{h}{4\pi m}$

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Work Sheet - 12

Complete the table :

Species	Electronic configuration	Unpaired electron	Spin only magnetic moment
${}_6\text{C}$			
${}_8\text{O}^{2-}$			
${}_{21}\text{Sc}$			
${}_{26}\text{Fe}^{2+}$			
${}_{24}\text{Cr}$			
${}_{29}\text{Cu}$			

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Work Sheet - 13

1. Complete the table :

Orbital	Angular node	Radial node	Total node	Orbital angular momentum
2s				
2p				
4p				
1s				
4f				
6s				

2. Mark valid set of quantum number

n	l	m	s	Mark (T / F)
3	2	2	+1/2	
2	0	0	-1/2	
4	2	-1	0	
4	3	4	-1/2	

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Work Sheet - 14

1. Complete the table :

S.No.	Properties	Alpha	Beta	Gamma
1.	Nature	Fast moving He nuclei	First moving electrons	High energy radiations
2.	Notation			γ or ${}^0_0\gamma$
3.	Charge	2 unit (+ ve)		
4.	Velocity		33% to 90% of light	
5.	Nature of path		Crooked	Waves
6.	Mass g/particle	6.65×10^{-24}		
7.	Electrical or magnetic field's influence			

2. Complete the table :

Parent element	${}_{90}\text{Th}^{232}$	${}_{94}\text{Pu}^{241}$	${}_{92}\text{U}^{238}$	${}_{92}\text{U}^{235}$
Name of series	Thorium (Natural)	Neptunium (Artificial)	Uranium (Natural)	Actinium (Natural)
End product	${}_{82}\text{Pb}^{208}$	${}_{83}\text{Bi}^{209}$	${}_{82}\text{Pb}^{206}$	
Number of lost particles				$\alpha = 7$ $\beta = 4$